

Thursday Commissioning Simulation — Instructor Pack

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Cyberrange*

GNG 3100 · SYS 5186

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Confidential — instructor and simulator operator only.

This pack covers the Thursday simulation day. Teams swap operator/ observer roles relative to Wednesday. Day target is **First Light** on the multispectral imager.

1. Day at a glance

When	What	Mode
09:00 – 09:30	Standup, scenario brief to both teams	Instructor-led
09:30 – 12:00	Pass-set 1 — Team A operates two compressed passes; Team B observes	Commissioning sim
12:00 – 13:00	Lunch + observer prep	—
13:00 – 13:15	Reset + brief for Pass-set 2	Instructor-led
13:15 – 15:45	Pass-set 2 — Team B operates two compressed passes; Team A observes	Commissioning sim
15:45 – 17:00	Joint debrief, observer presentations, anomaly tickets	Plenary + individual

Each pass-set covers ~4 simulated ground passes in 2.5 h real time using a compressed-time multiplier (3×–5× between passes, 1× during passes). The simulator handles the speed switching.

2. Scenario configuration

- **Breakpoint:** BP-3 — Commissioning Day Start (NEW; spec in `breakpoint_spec.md`).
- **Phase at start:** 4 (LEOP-complete).
- **Initial visibility:** Iqaluit, AOS at scenario T+0.
- **Planned passes during shift:** 4 (Iqaluit, Troll, Iqaluit, Troll — actual visibility depends on orbit).
- **Speed multiplier:** instructor-controlled; defaults to 3× between passes, 1× during passes.

[Claude Code: confirm the simulator’s compressed-time CLI options.]

3. Procedure expectations

The team is expected to drive commissioning in approximately this order across the four passes:

Pass 1. COM-001 EPS Checkout · COM-002 TCS Verification · COM-005 AOCS Mode Transitions (NADIR) · COM-006 TTC Link Verification.

Pass 2. COM-007 OBDH Checkout · COM-008 FDIR Configuration (load nominal table).

Pass 3. COM-009 Payload Power On · COM-010 FPA Cooler Activation (this runs through the eclipse and into pass 4 — heat sink during eclipse, cool-down on the day side).

Pass 4. COM-011 Payload Calibration · **COM-012 First Light** — acquire one image of an ocean-current target listed in `planning/imaging_targets.yaml`.

End-of-shift target state: phase 5+ (NOMINAL ready), payload calibrated, one usable image acquired.

4. Anomaly injection menu

Pick one primary and one optional secondary per team. Injection files live in `~/SpaceMissionSimulation/configs/eosat1/scenarios/`.

Severity	Scenario file	What students see	Recovery
Easy	eps_overcurrent.yaml (line= payload)	Overcurrent on payload power-on	CTG-012 Overcurrent Response
Easy	gs_rf_degradation.yaml	BER climbs mid-pass	CTG-014 BER Anomaly
Medium	fpa_overtemp.yaml	FPA cooler can't reach -60 °C	CTG-004 Thermal Exceedance + payload safe
Medium	payload_corrupt_image.yaml	First image returns corrupt	CTG-015 Corrupted Image Recovery
Hard	aocs_sensor_cascade.yaml	Star tracker degrades; pointing budget broken	CTG-008 Star Tracker Failure → CTG-002 + defer First Light
Hard	eps_progressive_load_shed.yaml	Battery degradation forces progressive load-shed during cooler activation	CTG-001 Under-Voltage Load Shed
Very hard	aocs_wheel_failure.yaml followed by transponder_failure.yaml	Wheel locks late in pass 3, then transponder fails next pass — multi-anomaly response	CTG-007 → CTG-003

[Claude Code: produce one cue card per scenario, same template as LEOP day \$4.]

5. Pass operator-side criteria

Criterion	Pass-set 1 (Team A)	Pass-set 2 (Team B)
Phase advanced to COMMISSIONING (5)	✓	✓
EPS checkout complete with no faults	✓	✓
AOCS NADIR mode achieved, pointing < 0.5°	✓	✓
FDIR nominal table loaded	✓	✓
Payload powered ON	✓	✓
FPA cooler at temperature (≤ -55 °C)	✓	✓
First Light image acquired (or formally deferred)	✓	✓
Anomaly tickets filed for all injections	✓	✓

The “formally deferred” option is genuine: if the team correctly diagnoses that pointing is insufficient and chooses to defer imaging, they pass the criterion. If they image anyway and produce garbage, they fail.

6. Coaching during the shift

[Claude Code: copy §6 of `sim_leop_instructor.md` and adapt for commissioning specifics. Add: watch the team’s handover quality between passes, since pass-to-pass continuity is harder than within a single pass.]

7. Debrief

[Claude Code: same structure as `sim_leop_instructor.md` §7, with the difference that observer worksheets used today are Team and Space usage (the other two were used Wednesday).]

8. Safety, red lines, references

[Claude Code: copy from `sim_leop_instructor.md` §§8 – 9 and add the Commissioning-day-specific files:

- `scenarios/commissioning_day_start.yaml` (NEW)
- `procedures/commissioning/*.md`
- The `mcs/role_analysis/payload_ops_role.md` (today is heavy on payload ops).]